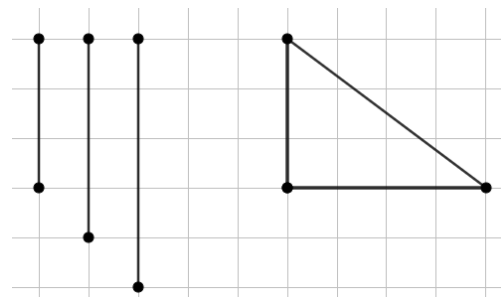
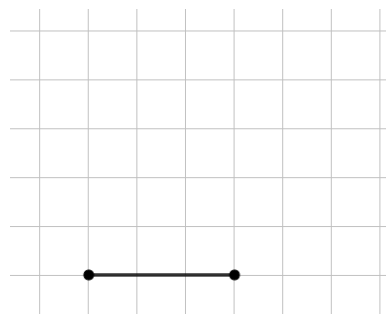


Three line segments and a triangle are shown.

1. Compare the triangle that was constructed with the three line segments. Describe any similarities or differences.



2. One of the line segments is shown on the grid. Draw a different triangle that uses the other two line segments.



3. Describe how you could determine if the angles of the original triangle and the angles of triangle you drew in question 2 are congruent.

4. Complete the statement.

- ☐ Any
☐ Some
☐ No

triangles created using the three line segments are congruent to

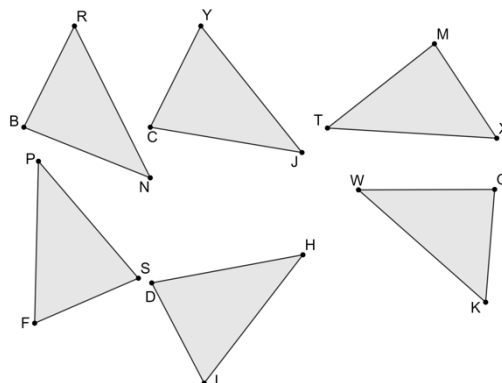
the original triangle.

Six triangles are shown.

Use patty paper, a blank sheet of paper, a ruler, and/or a protractor to help answer the following questions.

5. Which triangles have a side that has the same length as $\overline{BR}$?

6. Which triangles have a side that has the same length as $\overline{BN}$?



7. Which triangles have a side that has the same length as $\overline{RN}$?

8. Complete the statements.

All of the triangles that have a side that have the same length as $\overline{BR}$

☐ are
☐ are not

congruent to $\triangle RBN$. All of the triangles that have two

sides that have the same length as $\overline{BR}$ and $\overline{BN}$

☐ are
☐ are not

congruent to $\triangle RBN$. All of triangles that have three sides that have

the same length as $\overline{BR}$, $\overline{BN}$, and $\overline{RN}$

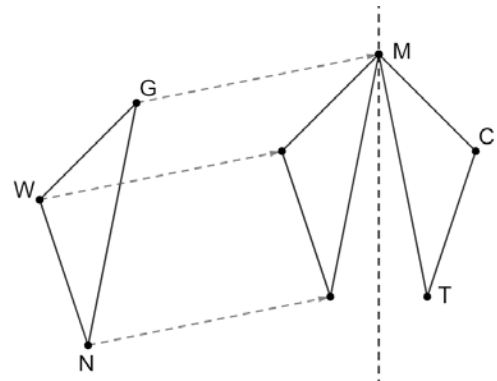
☐ are
☐ are not

congruent to $\triangle RBN$.

A series of transformations shows how $\overline{GW}$, $\overline{GN}$, and $\overline{NW}$ were translated then reflected across a line.

9. Select the reason the transformations that were used ensure the length of each line segment was preserved.

- ☐ The transformations included a dilation.
- ☐ The transformations were rigid motions.
- ☐ The transformations did not include a rotation.



10. Complete the statements.

The definition of congruence in terms of _____

ensures that the _____ are congruent. Because all of the

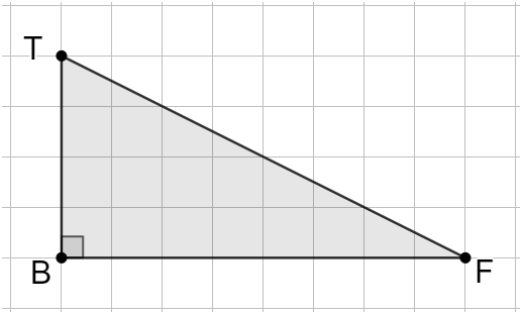
_____ of $\triangle GNW$ are _____ to all of the corresponding

_____ of $\triangle MTC$, the two triangles are _____. This shows that

the _____ congruence postulate is true.

Triangle TBF is shown on a grid.

Construct $\triangle HKM$ so that $\overline{TB} \cong \overline{KM}$, $\angle TBF \cong \angle HMK$, and $\overline{FB} \cong \overline{HM}$ using the given steps, questions 1 through 4.

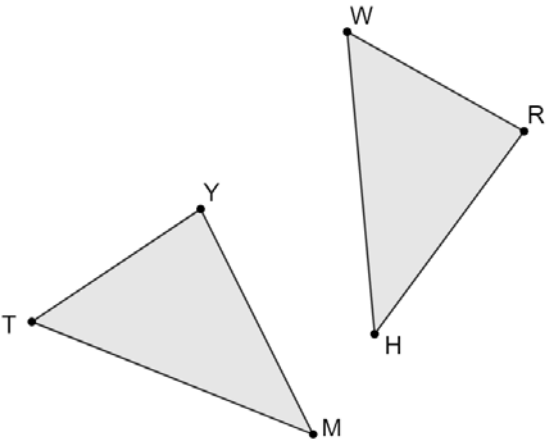


Steps	Construction
1. Draw $\overline{KM}$ on the grid so that $KM = TB$.	
2. Draw $\angle HMK$ on the grid so that $m\angle HMK = m\angle TBF$.	
3. Draw $\overline{HM}$ on the grid so that $HM = FB$.	
4. Draw $\overline{KH}$.	

5. Describe how to use geometric tools to determine if $\overline{TF} \cong \overline{KH}$, $\angle BFT \cong \angle MHK$, and $\angle FTB \cong \angle HKM$.
6. Since $KM = TB$, $m\angle HMK = m\angle TBF$, and $HM = FB$, then by the definition of congruence $\overline{TB} \cong \overline{KM}$, $\angle TBF \cong \angle HMK$, and $\overline{FB} \cong \overline{HM}$. Which congruence postulate or theorem does the construction illustrate?

Triangles YMT and WRH are shown.

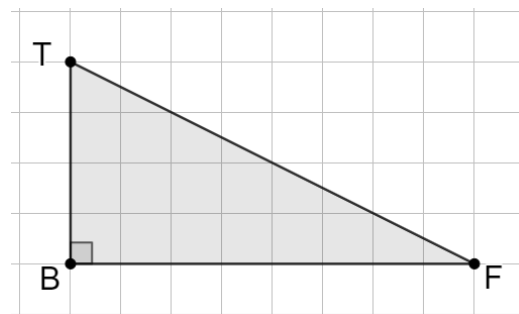
Determine if the given information is sufficient to determine whether the triangles are congruent. If so, state the congruence theorem or postulate.



	Given Information	Are the Triangles Congruent?	Theorem or Postulate Used
7.	$\overline{TY} \cong \overline{WR}, \overline{YM} \cong \overline{HR}, \overline{TM} \cong \overline{WH}$		
8.	$\overline{YM} \cong \overline{HR}, \angle YTM \cong \angle RWH, \overline{TM} \cong \overline{WH}$		
9.	$\angle YTM \cong \angle RWH, \overline{TY} \cong \overline{WR}, \overline{TM} \cong \overline{WH}$		
10.	$\angle MYT \cong \angle HRW, \overline{YM} \cong \overline{HR}, \angle YMT \cong \angle RHW$		

Triangle TBF is shown on a grid.

1. Draw $\overline{KW}$ on the grid so that $KW = BT$.
2. Draw $\overline{WN}$ on the grid so that $WN = BF$ and $\overline{WN}$ is perpendicular to $\overline{KW}$.
3. Draw $\overline{KN}$ using a straightedge.



4. Complete the statements.

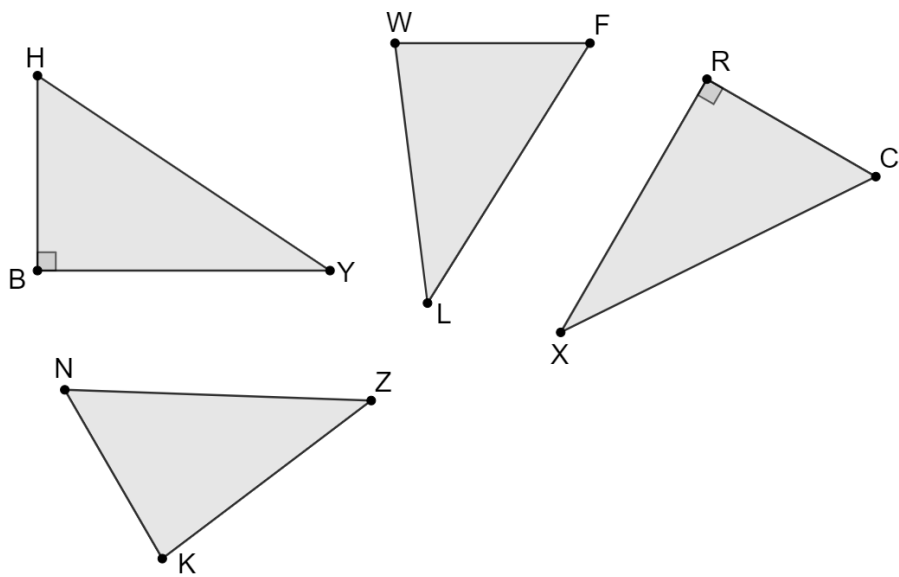
By the definition of congruence, since $KW = BT$ then $\overline{KW} \cong \overline{BT}$ and since $WN = BF$ then $\overline{WN} \cong \overline{BF}$. Since $m\angle TBF = 90^\circ$ and $m\angle KWN = 90^\circ$, then by applying the transitive property and the definition of congruence, then $\angle TBF \cong \angle KWN$. The construction of $\triangle KWN$ uses

- ☐ ASA
- ☐ SAS
- ☐ SSS

to show that $\triangle TBF \cong \triangle$ _____.

Because both triangles are right triangles the triangles could also be shown as congruent by showing that $\overline{KW} \cong \overline{BT}$ and that _____ $\cong$ _____ or that $\overline{WN} \cong \overline{BF}$ and that _____ $\cong$ _____.

Triangles KZN , BYH , LWF , and XRC are shown.



Determine if the given information is sufficient to determine whether the triangles are congruent. If so, state the congruence theorem or postulate.

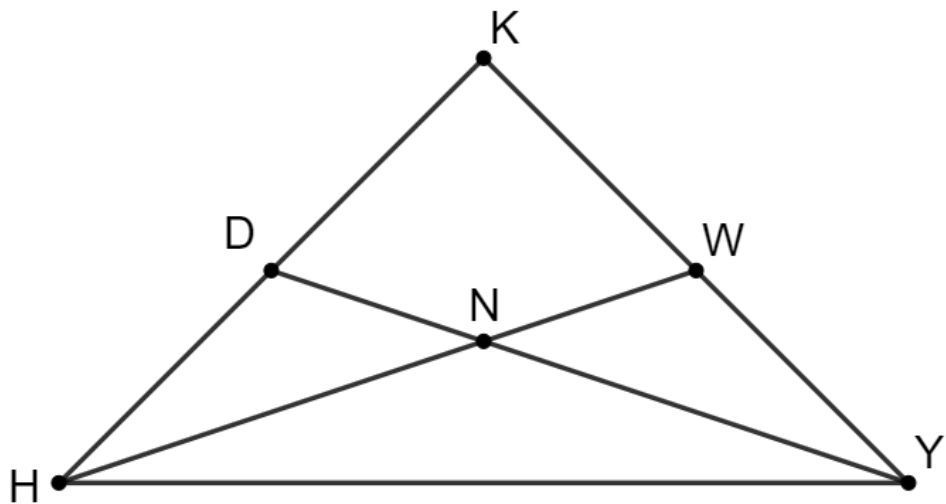
	Given Information	Are the Triangles Congruent?	Theorem or Postulate Used
5.	$\overline{BY} \cong \overline{XR}$, $\overline{YH} \cong \overline{CX}$		
6.	$\angle NZK \cong \angle WLF$, $\overline{NK} \cong \overline{FW}$, $\angle ZKN \cong \angle FWL$		
7.	$\overline{HB} \cong \overline{RC}$, $\overline{BY} \cong \overline{XR}$, $\overline{HY} \cong \overline{CX}$		
8.	$\angle WFL \cong \angle ZNK$, $\overline{WF} \cong \overline{NK}$, $\angle NZK \cong \angle WLF$		
9.	$\angle ZKN \cong \angle FWL$, $\overline{NZ} \cong \overline{FL}$, $\overline{KZ} \cong \overline{LW}$		
10.	$\angle YBH \cong \angle CRX$, $\angle BHY \cong \angle XCR$, $\overline{HY} \cong \overline{CX}$		

Geometry
Unit 5
Lesson 4 Homework

Name _____
Date _____
Period _____

Triangle KYH is shown.

1. List all of the triangles in the diagram.



If $\triangle HYD \cong \triangle YHW$, complete the congruence statements.

2. $\overline{WY} \cong$

3. $\angle YDH \cong$

4. $\overline{WH} \cong$

5. $\angle WYH \cong$

Triangle TMF is shown.

If $\triangle TLF \cong \triangle MLF$, complete the congruence statements.

6. $\overline{FL} \cong$

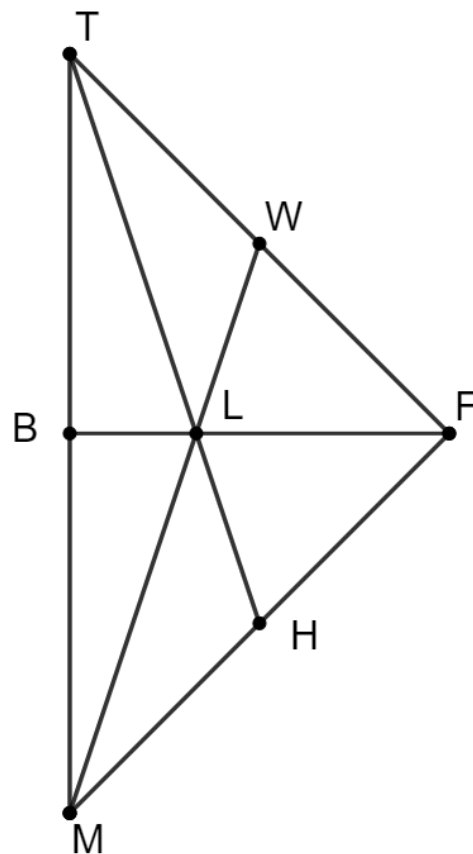
7. $\angle FMW \cong$

8. $\overline{FM} \cong$

If $\triangle FBT \cong \triangle FBM$, complete the congruence statements.

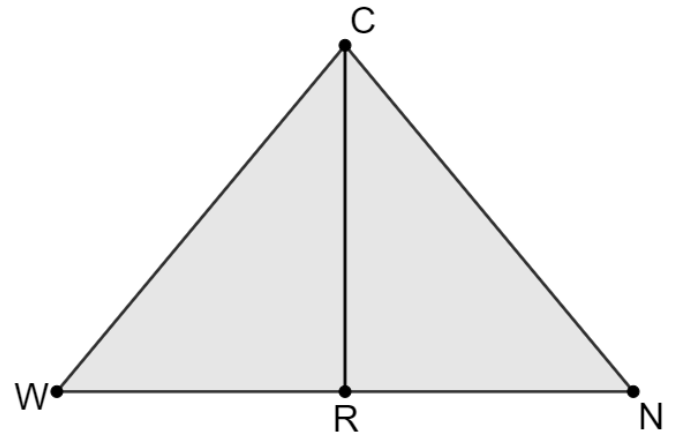
9. $\overline{TF} \cong$

10. $\angle FBT \cong$



Triangle CWN is shown.

1. Josh marked $\angle CRW$ with a right-angle symbol. Explain to Josh why he cannot mark $\angle CRW$ with a right-angle symbol by just looking at the figure.



2. Mark the diagram using the given information shown.

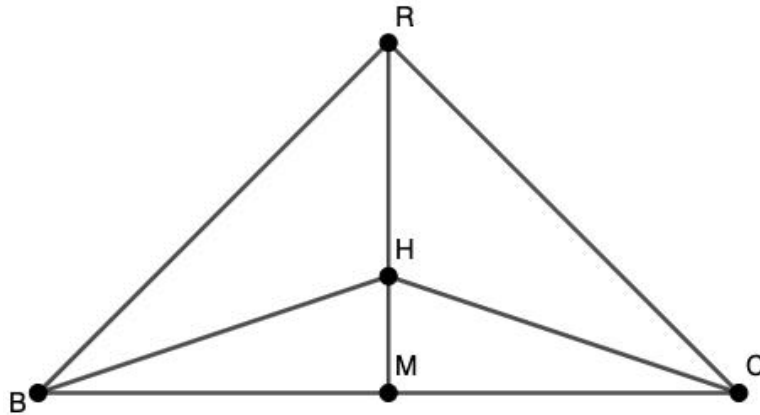
$\overline{CR}$ is the perpendicular bisector of $\overline{WN}$.

3. Which side(s) of triangles CRW and NRC are congruent because of the given information?
4. Which angle(s) of triangles CRW and NRC are congruent because of the given information?
5. Which side(s) or angle(s) of triangles CRW and NRC are congruent because of the reflexive property?
6. Using the given information, which, if any, of the triangle congruence theorems or postulates could be used to determine that the triangles are congruent.

☐ SSS☐ SAS☐ ASA☐ AAS☐ HL☐ none

Use the given information to answer questions 7-10.

Given: $\triangle BHM \cong \triangle CHM$



7. Mark the diagram with the congruent parts and angles knowing $\triangle BHM \cong \triangle CHM$.

8. What other information is needed to show $\triangle BRM$ and $\triangle CRM$ are congruent?

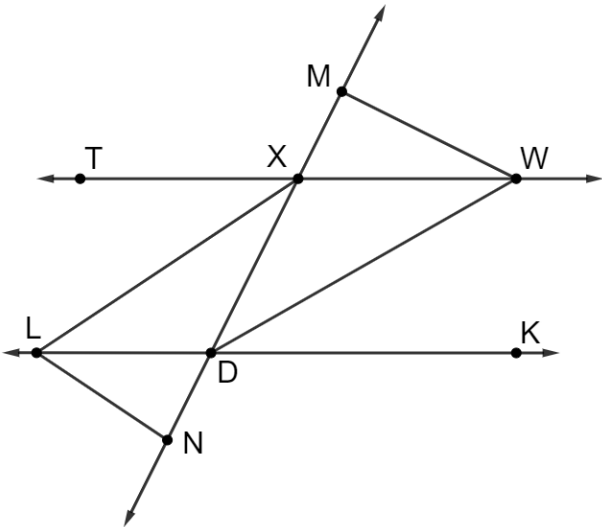
Complete statements.

9. $\overline{RM} \cong \overline{RM}$ because of the _____.

10. $\angle BMH \cong \angle CMH$ and $\overline{BM} \cong \overline{CM}$ by _____.

Therefore, $\triangle BRM \cong \triangle CRM$ by _____.

A diagram is shown where $\overline{TW} \parallel \overline{LK}$,
 $\overline{XW} \cong \overline{LD}$, and $\angle MWX \cong \angle NLD$.



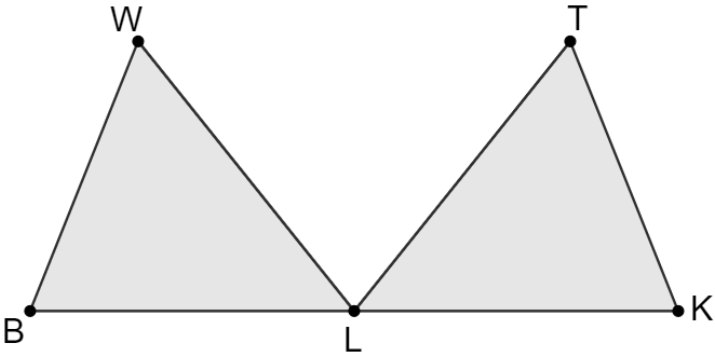
Use the given information to answer the questions about the incomplete proof that is shown.

Statement	Reason
1. $\overline{XW} \cong \overline{LD}$	1. Given
2. $\overline{TW} \parallel \overline{LK}$	2. Given
3.	3.
4.	4. Reflexive property
5. $\triangle XDL \cong \triangle \underline{\hspace{1cm}}$	5.

- What is the statement for line 3 of the proof?
- What is the reason for line 3 of the proof?
- What is the statement for line 4 of the proof?
- Complete the congruence statement for the statement of line 5 of the proof.
 $\triangle XDL \cong \triangle \underline{\hspace{1cm}}$
- What is the reason for line 5 of the proof?

A diagram is shown where point L is the midpoint of $\overline{BK}$, $\overline{LW} \cong \overline{LT}$, and $\angle WLB \cong \angle TLK$.

Use the given information to answer the questions about the incomplete proof that is shown.



Statement	Reason
1. $\overline{LW} \cong \overline{LT}$	1. Given
2. $\angle WLB \cong \angle TLK$	2.
3. L is the midpoint of $\overline{BK}$	3. Given
4.	4.
5. $\triangle WBL \cong \triangle \underline{\hspace{1cm}}$	5.

6. What is the reason for line 2 of the proof?
7. What is the statement for line 4 of the proof?
8. What is the reason for line 4 of the proof?
9. Complete the congruence statement for the statement of line 5 of the proof.

$$\triangle WBL \cong \triangle \underline{\hspace{1cm}}$$
10. What is the reason for line 5 of the proof?

Geometry

Unit 5

Lesson 7 Homework

Name _____

Date _____

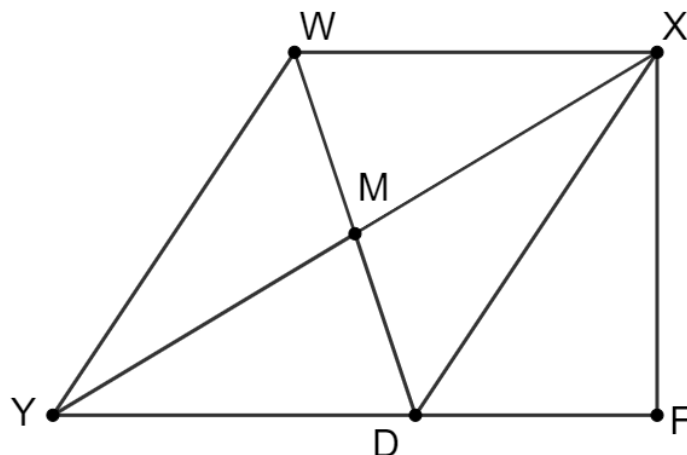
Period _____

Quadrilateral $WXYF$ is shown where M is the midpoint of $\overline{WD}$ and $\overline{WX} \parallel \overline{YF}$.

Given: M is the midpoint of $\overline{WD}$, $\overline{WX} \parallel \overline{YF}$.

Prove: $\overline{WX} \cong \overline{YD}$

Use the given information to answer the questions about the incomplete proof that is shown. Each empty box has the corresponding question number.



M is the midpoint of $\overline{WD}$

Given

Question 1

Question 2

$\overline{WX} \parallel \overline{YF}$

Given

Question 3

Question 4

Question 5

Question 6

Question 7

Question 8

Question 9

Question 10

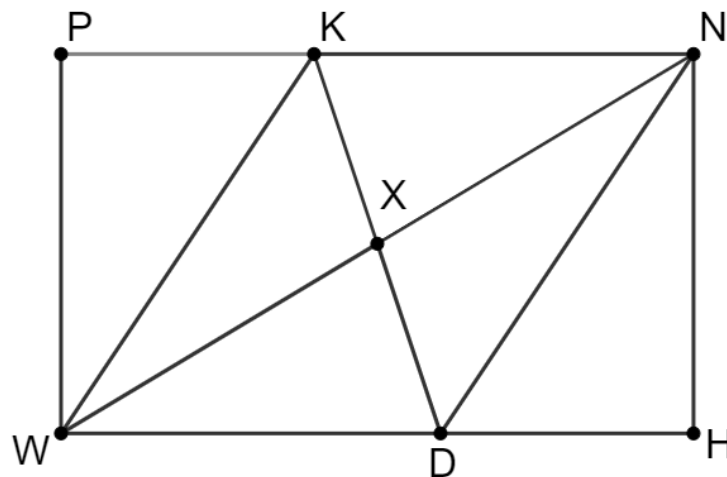
1. What is the statement that belongs in the box?
2. What is the reason for the statement?
3. What is the statement that belongs in the box?
4. What is the reason for the statement?
5. What is the statement that belongs in the box?
6. What is the reason for the statement?
7. What is the statement that belongs in the box?
8. What is the reason for the statement?
9. What is the statement that belongs in the box?
10. What is the reason for the statement?

Quadrilateral $PNHW$ is shown where $\overline{PN} \perp \overline{WP}$, $\overline{WH} \perp \overline{NH}$, and $\overline{PW} \cong \overline{NH}$.

Given: $\overline{PN} \perp \overline{WP}$, $\overline{WH} \perp \overline{NH}$, $\overline{PW} \cong \overline{NH}$

Prove: $\overline{PN} \cong \overline{WH}$

Use the given information to answer the questions about the incomplete proof that is shown.



Statement	Reason
1. $\overline{PN} \perp \overline{WP}$, $\overline{WH} \perp \overline{NH}$	1. Given
2. $\angle WPN$ and $\angle NHW$ are right angles	2.
3.	3. definition of right angles
4.	4.
5.	5. definition of congruence
6. $\overline{PW} \cong \overline{NH}$	6. Given
7.	7.
8.	8.
9. $\overline{PN} \cong \overline{WH}$	9.

1. What is the reason for line 2 of the proof?

2. What is the statement for line 3 of the proof?

3. What is the statement for line 4 of the proof?

4. What is the reason for line 4 of the proof?

5. What is the statement for line 5 of the proof?

6. What is the statement for line 7 of the proof?

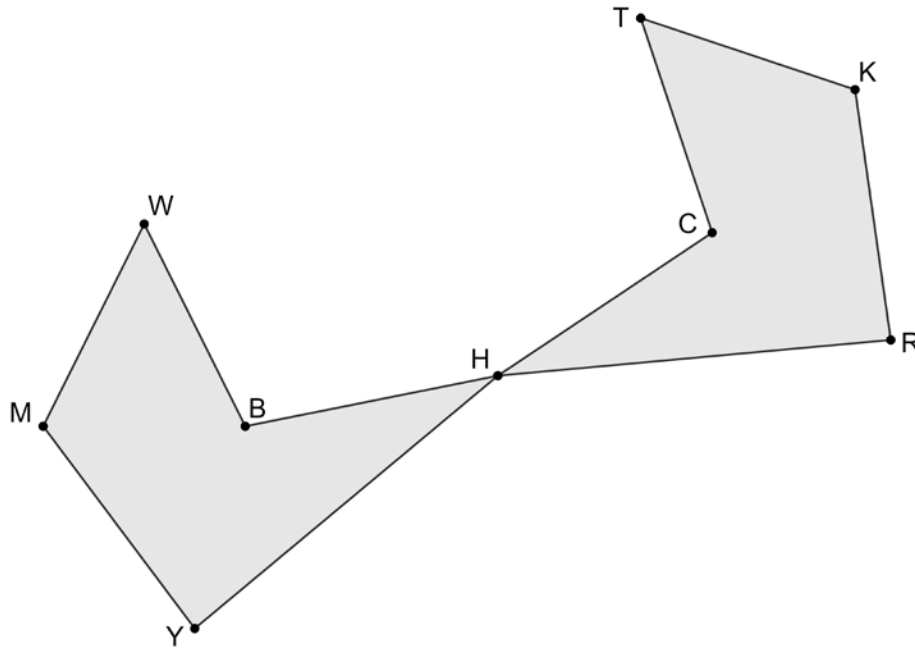
7. What is the reason for line 7 of the proof?

8. What is the statement for line 8 of the proof?

9. What is the reason for line 8 of the proof?

10. What is the reason for line 9 of the proof?

Congruent polygons $WBHYM$ and $TCHRK$ are shown where $\overline{TC} \cong \overline{TK}$.



1. Determine the length of $\overline{TC}$ if $TC = 8x - 11$ and $TK = 3x + 24$.
2. Determine the value of y when the perimeter of $WBHYM$ is 236 units and $HR = 7y - 6$, $KR = 5y - 11$, and $BH = 4y + 3$.
3. What is the length of $\overline{HR}$?
4. What is the length of $\overline{MY}$?
5. What is the length of $\overline{HC}$?

Some of the angle measures of $WBHYM$ and $TCHRK$ are shown.

$$m\angle WMY = (8k - 19)^\circ$$

$$m\angle HRK = (5k + 2)^\circ$$

$$m\angle KTC = (3k + 2)^\circ$$

$$m\angle RHC = (2k - 6)^\circ$$

$$m\angle WBH = 9(m\angle RHC) + 3^\circ$$

$$m\angle RKT = (5k + 32)^\circ$$

6. Determine the value of k .

Determine each angle measure.

7. $m\angle BWM = \underline{\hspace{2cm}}$

8. $m\angle WBH = \underline{\hspace{2cm}}$

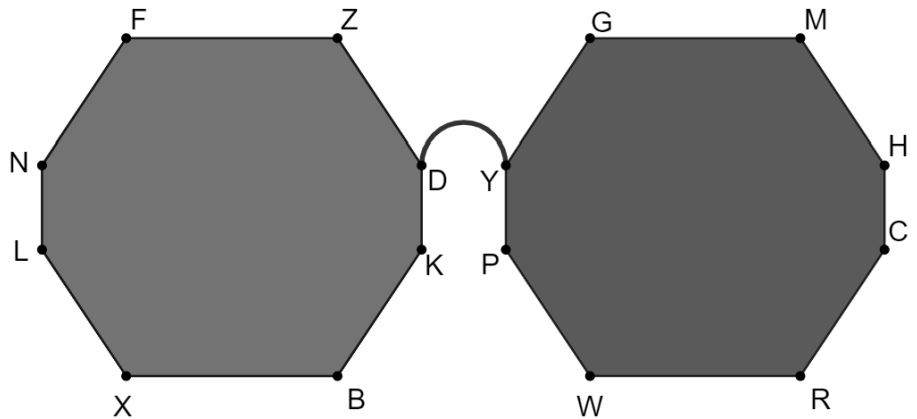
9. $m\angle MYH = \underline{\hspace{2cm}}$

10. $m\angle TCH = \underline{\hspace{2cm}}$

Geometry
Unit 5
Lesson 10 Homework

Name _____
Date _____
Period _____

Minnie is designing sunglasses. She uses a computer drawing program to create a diagram of the lenses. Her drawing is shown with congruent polygons $FZDKBXLN$ and $GMHCRWPY$ representing the lenses of the sunglasses.



In polygon $FZDKBXLN$, $\overline{FZ} \cong \overline{XB}$, $\overline{NL} \cong \overline{DK}$, and $\overline{DZ} \cong \overline{KB} \cong \overline{LX} \cong \overline{NF}$. The perimeter of polygon $GMHCRWPY$ is 56.8 millimeters (mm).

Some of the side measures of $GMHCRWPY$ are shown.

$$PW = 6x - 2.4$$

$$HC = 2(3x - 2.8)$$

$$WR = 5x + 2$$

- Write an equation that can be used to determine the value of x .
- Determine the value of x .
- Determine the length of side GM .
- Determine the length of side BK .
- Determine the length of side NL .

Some of the angle measures of $FZDKBXLN$ are shown.

$$m\angle BXL = (4k - 8)^\circ$$

$$m\angle DZF = (3k + 25)^\circ$$

$$m\angle XLN = (5k - 19)^\circ$$

$$m\angle XBK = (5k - 41)^\circ$$

$$m\angle NFZ = m\angle DZF$$

$$m\angle DKB = (3k + 47)^\circ$$

$$m\angle XLN = m\angle LNF$$

$$m\angle ZDK = m\angle DZF + 22^\circ$$

6. Given that $m\angle NFZ + m\angle FZD + m\angle MHC = 394^\circ$, determine the value of k .

Determine the measure of each angle.

7. $m\angle BXL = \underline{\hspace{2cm}}$

8. $m\angle WRC = \underline{\hspace{2cm}}$

9. $m\angle PYG = \underline{\hspace{2cm}}$

10. $m\angle RCH + m\angle CHM = \underline{\hspace{2cm}}$